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Joint variational Bayesian based localization estimation algorithm on distributed gas source sensor network

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ABSTRACT

According to the non-linear and unknown diffusion distribution model characteristics of gas leakage diffusion in the real environment, a distributed joint optimal estimation algorithm for gas source localization and parameters estimation was proposed in sensor networks. Firstly, the gas source localization framework was constructed based on the compressed sensing theory, Secondly, the joint sparse estimation method of the state distribution and unknown parameters of gas diffusive source is proposed based on the variational Bayesian inference algorithm. In which, an adaptive grid division strategy is given to improve the accuracy and performance of the joint estimation method and to balance the relationship between energy consumption and network resources. Finally, simulation results show that the proposed algorithm could effectively achieve a joint estimation of the diffusive source state and parameters, which could achieve higher estimation accuracy in a shorter time and meet the real-time requirements in complex environment compared with the traditional CS method.

1. Introduction

With the rapid development of embedded system, distributed computing and wireless communication technology, sensor networks has attracted more and more attention in the field of target source localization and tracking, which has been widely used in many fields such as the radar detection [1], sonar localization [2], medical image [3], pollution source localization [4] and so on.

In recent years, the hazardous chemical leakage source localization in sensor network has become a hotspot in the environmental monitoring research field [5]. With the cooperation of sensor nodes, the environmental measurements and information interaction could be get and the target leakage source location could be accurately predicted, which would provide a strong support for pollution prevention and accident rescue.

The distributed gas leakage source localization in sensor networks can be described as an iterative estimation problem of unknown parameters of gas leakage source in environment, or a inversion problem of non-linear dynamic gas diffusion distribution model [6–9], which is mainly solved by the joint estimation method based on Bayesian inference and Markov random probability process theory [10].

In the traditional sensor network target source localization methods, the system state and observation model are usually assumed to be priori known and it could be implemented by the approximate linear filtering method, such as the iterative EKF [11], the iterative extended Kalman filter [12] and the particle filter (PF) [13,14]. As for the multi-target source localization and parameters identification with the unknown structure of system model, the distributed iterative joint optimization estimation method based on Bayesian inference theory is generally used. In which, the EM and variational Bayes Inference has become the main method. Compared to the EM algorithm, the variational Bayesian inference [15] is especially for the target state and unknown parameter estimation of multi-target source location system in the distributed sensor network [16].

In many target localization applications with sensor network, the interested target sources can be regarded as relative sparse to the monitoring background, or the information sampling rate of sensor network is limited. In order to obtain accurate positioning performance, large amount of redundant environment information should be sampled. The massive information collection and transmission in sensor networks will consume many resources and reduce network life. There is a great imbalance between the sparse characteristics of the target source and the massive sampling data or sampling rate needed for source localization. It is a key problem to reduce the data transmission and resource consumption effectively in sensor networks. Recently, the emerging theory of compressed sensing (CS) [17] provides a new choice to solve this imbalance problem, which could be used to improve location performance greatly by reducing the amount of sampling data and information transmission with the sparse characteristics of multi-target sources in sensor networks.

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Considering the sparsity of the target sources in the monitor environment, the CS multi-target source localization methods usually divide the monitoring area into grids firstly. In the literature [18], the monitoring area is divided into several grids and the multi-target source localization problem was transformed into a K sparse signal reconstruction problem based on the CS theory in the sensor network. The target source vector was reconstructed from the compressed observation vector by the cluster head node using L1 optimization algorithm. Considering the high computational complexity of L1 optimization reconstruction algorithm, in the paper [19], the greedy matching pursuit (GMP) algorithm was used to realize multi-target source localization, but the accuracy error of the localization algorithm need to be further improved.

In [20], an improved GMP algorithm was proposed, in which, the greedy matching residual optimization algorithm was used for the K sparse signal reconstruction and the positioning accuracy improvement. Although the complexity of sparse reconstruction algorithm based on greedy matching pursuit is lower, it needed prior signal sparsity or the number of targets. However, in practical applications, the number of targets is usually unknown and the amount of sampling data for the signal compressed sensing reconstruction could hardly be determined. Therefore, how to determine the optimal number of observations is one of the key problems, which could overcome the insufficient sampling or oversampling problems in the GMP algorithms.

In order to accurately identify and localize the multi-target sources, it is usually necessary to increase the grid division density or the number of grids, which will greatly increase the complexity of the sparse reconstruction model or the computation of the reconstruction algorithm. And will increase the positioning error of the target source. In the literature [21], an adaptive grid division method was proposed for the multi-targets source localization problem with the Sparse Bayesian learning algorithm. Simulation results show that the adaptive grids division method could get a lower localization error than the fixed grid division method.

In [22], a multi-target location algorithm with hierarchical grids division method is given: in the coarse localization stage, the initial candidate target grid is reconstructed by L1 optimization and in the fine location stage, each initial candidate grid was divided into four parts in turn, and the exact location of the target is determined in the smaller sub grid.

In this paper, aiming at the joint distribution estimation problem of the non-linear gas diffusion distribution state and the unknown source parameters in the complex environment, a hierarchical gas leakage source localization algorithm with adaptive grid division method is proposed, in which, the environment was mainly divided with two scales with the global large-size and the local small size. In the global grid division and localization stage, the joint estimation of the sparse gas leakage source distribution state and the initial candidate grids within targets were realized based on variational Bayesian inference algorithm, in which, the system state model was described with Gaussian mixture model to approximate the posterior probability distribution. In the local adaptive small-scale grid division and localization stage, the initial candidate grid is adaptively divided with the estimator errors, and the precise position of the target source and the sparse reconstruction of the related unknown parameters in the initial grid are realized by the parallel variational Bayesian inference in different cluster sensor nodes.

The structure of the paper is as follows: in Section 2, the compressed sensing framework and system model for the multi-target source localization problem are described, and In Section 3, the joint estimation algorithm of gas distribution state and source unknown parameters is proposed based on variational Bayes inference method. In Section 4, the adaptive grids division method for the gas leakage target localization algorithm is given. In Section 5, the algorithm is verified with simulations and the results are analyzed. Section 6 is the conclusion.

2. System model and compressed sensing framework for gas source localization

Assuming M sensor nodes with known locations and K target sources with unknown locations are randomly deployed in the monitoring field. As for the compressed sensing multi-target source localization method, the monitoring field should be divided into grids and each target should only be appeared in the center of some grid.

The gas diffusion in the environment is a highly nonlinear stochastic process and it is usually hardly to establish a diffusion model in the global monitored area accurately [23]. However, for the local area, the relatively approximate gas diffusion model can be deduced by the turbulence diffusion theory. In this paper, it is assumed that the gas leakage substance diffusion distribution in the local grid area covered by a single cluster sensor network conforms to a gas diffusion model given in [24], which is described as follows:

$$c(\boldsymbol{r}_m, t) = \frac{q}{2\pi\rho}\frac{1}{\|\boldsymbol{r}_m - \boldsymbol{r}_s\|}\exp\left[-\frac{U}{2K}\left(\|\boldsymbol{r}_m - \boldsymbol{r}_s\|\right)\right] \tag{1}$$

where $\boldsymbol{r}_s = (x_s, y_s)$ is the coordinate vector of gas source location, $c(\boldsymbol{r}_m, t)$ is the gas diffusion concentrations at the sensor node location $\boldsymbol{r}_m = (x_m, y_m)$, $m = 1, 2, \ldots, M$, ρ is the total number of sensor nodes in the network); q is the gas release rate; ρ is the turbulent diffusion coefficient; $c(\boldsymbol{r}_m, t)$ is the wind speed.

With the grid division of the monitoring area, it can be transformed $c(\boldsymbol{r}_m, t)$ into an approximate solution of the discrete distribution state in different grid areas, in which, $c(\boldsymbol{r}_m, t)$ will be approximately described by the weighted series:

$$c(\boldsymbol{r}_m, t)\sum_{n=1}^{N} a_n(t)\cdot \boldsymbol{X}_n(t) \tag{2}$$

where $a_n(t)$ is the state conversion function, $\boldsymbol{X}_n(t)$ is the distribution state of gas sources in different grid locations, N is the total number of grids in the monitoring area. The distribution state of K targets in the whole monitoring area at the t clock period can be described by vectors $\boldsymbol{X}(t) = \left[\boldsymbol{X}_n(t),\ n = 1, 2, \ldots, N\right]$. The value of the $\boldsymbol{X}_n(t)$ is that there are targets in the nth grid $\boldsymbol{X}_n(t) = 1$ else $\boldsymbol{X}_n(t) = 0$. At the same time, in the t period, the observation vector of the sensor node with coordinate position $\boldsymbol{r}_m$ is defined as $\boldsymbol{Y}_n(t)$, and its observation model is described as:

$$\boldsymbol{Y}_n(t) = \boldsymbol{A}_n(t)\cdot \boldsymbol{X}_n(t) + \xi_n(t) \tag{3}$$

where $\boldsymbol{\xi}_n(t)$ is the observation noise and obeys Gaussian white noise distribution with mean $\boldsymbol{u}$ and the variance $\boldsymbol{C}$. $\boldsymbol{A}_n(t) = [a_n^1(t), a_n^2(t), \ldots, a_n^M(t)]^T$ is the observation matrix, $a_n^m(t)$, $m = 1, 2, \ldots, M$ is the state conversion function, and M is the number of sampling points or nodes in the observation sequence of time dimension.

In the compressed sensing theory, the formula (3) is usually assumed to be a sparse representation model and $\boldsymbol{A}_n(t) = [a_n^1(t), a_n^2(t), \ldots, a_n^M(t)]^T$ to be a compressed sensing matrix. The high-dimensional signals can be projected into the low-dimensional compressed space with $\boldsymbol{A}_n(t) = [a_n^1(t), a_n^2(t), \ldots, a_n^M(t)]^T$, and then the original signals can be recovered with high probability by solving the following optimized constraint problems:

$$\hat{\boldsymbol{X}}_n(t) = \arg\min\left\|\boldsymbol{X}_n(t)\right\|_0 \ \ s.t.\left\|\boldsymbol{Y}_n(t) - \boldsymbol{A}_n(t)\boldsymbol{X}_n(t)\right\|_2^2 = |\boldsymbol{C}|^2 \tag{4}$$

where $\left\|\boldsymbol{Y}_n(t) - \boldsymbol{A}_n(t)\boldsymbol{X}_n(t)\right\|_2^2$ is a L_2 norm. Because $M < N$, the above inverse process is a *NP* hard problem. Generally, it is impossible to get the definite solution of the system, but due to the existence sparsity of $\boldsymbol{X}_n(t)$, it is expected to get the definite solution with the linear programming method. In [25], it points out that the L0 optimization problem can be approximated by an L_1 optimization problem under certain conditions as follow:

$$\hat{\boldsymbol{X}}_n(t) = \arg\min\left\|\boldsymbol{X}_n(t)\right\|_1 \ \ s.t.\left\|\boldsymbol{Y}_n(t) - \boldsymbol{A}_n(t)\boldsymbol{X}_n(t)\right\|_2^2 \le |\boldsymbol{C}|^2 \tag{5}$$

where $\|\boldsymbol{X}_n(t)\|_1$ represents the L_1 norm of signal. The equivalent condition of formula (4) and formula (5) is that the perception matrix satisfies the condition of restricted isometric property (RIP) [25].

$$(1-\delta)\|\boldsymbol{X}_n(t)\|_2^2 \le \|\boldsymbol{A}_n(t)\boldsymbol{X}_n(t)\|_2^2 \le (1+\delta)\|\boldsymbol{X}_n(t)\|_2^2 \tag{6}$$

where δ is a small constant. Generally, the sparsity of signals is unknown. The optimization problem in formula (5) can be transformed into a constrained quadratic programming problem

$$\hat{\boldsymbol{X}}_n(t) = \arg\min\left(\|\boldsymbol{Y}_n(t) - \boldsymbol{A}_n(t)\boldsymbol{X}_n(t)\|_2^2 + \eta\|\boldsymbol{X}_n(t)\|_1\right) \tag{7}$$

where η is the optimization balance parameters, and the formula (7) can generally be solved by common optimization algorithms such as orthogonal matching pursuit (OMP) algorithm [26], sparse Bayesian learning (SBL) algorithm [27] and so on.

In the compressed sensing gas release source location problem, M sensor nodes are mainly used to obtain the observation vector $\boldsymbol{Y}_n(t)$ based on compressed sensing matrix $\boldsymbol{A}_n(t)$ and it will be transmitted to the cluster node. The K sparse vector $\boldsymbol{X}(t)$ is reconstructed with the cluster node to realize the number and location estimation of the target source. So, the number of non-zero elements in $\boldsymbol{X}(t)$ is the number of K targets and the corresponding grid position is K Target locations.

3. Gas source state and parameters joint estimation with sequential variational Bayesian inference algorithm

The gas leakage source localization problem usually can be attributed to a uncertain gas source distribution state and unknown parameter identification problem with unknown system model. In this paper, the system state distribution was premise described with Gaussian mixture models, and a joint variational Bayesian estimation algorithm is proposed for the target state reconstruction and the source unknown parameters estimation, which could solve the problem of the unknown variables estimation under the high-dimensional complex distribution.

3.1. Gaussian mixture model

For the system with unknown parameters of gas diffusion model, its posterior probability distribution function cannot be described directly with a single Gaussian distribution model. The linear weighted combination of multiple Gaussian distributions is usually used for the observation system model with unknown parameters vector $\boldsymbol{\Theta}$:

$$\boldsymbol{A}_n(t)\cdot\boldsymbol{X}_n(t)\sum_{g=1}^{G}\lambda_g\cdot N\left(\boldsymbol{X}_n(t)-\hat{\boldsymbol{X}}_{ng}(t),\boldsymbol{C}_g^x\right) \tag{8}$$

where G is the number of independent Gaussian probability density function components, and λ_g is the weight of each independent Gaussian function, which obeys the following conditions:

$$\sum_{g=1}^{G}\lambda_g = 1, \quad \lambda_g \ge 0 \tag{9}$$

Assuming that the probability density function of the finite Gaussian mixture is the same, the probability density of the g mixture follows:

$$\mathbb{N}\left(\boldsymbol{X}_n(t)-\hat{\boldsymbol{X}}_{ng}(t)\right) \propto \frac{1}{(2\pi)\left|\boldsymbol{C}_g^x\right|^{1/2}} e^{-1/2\left(\boldsymbol{X}_n(t)-\hat{\boldsymbol{X}}_{ng}(t)\right)^T\left(\boldsymbol{C}_g^X\right)^{-1}\left(\boldsymbol{X}_n(t)-\hat{\boldsymbol{X}}_{ng}(t)\right)} \tag{10}$$

where $\left(\hat{\boldsymbol{X}}_{ng}(t),\boldsymbol{C}_g^x\right)$ is the ith independent normal Gaussian component, $\hat{\boldsymbol{X}}_{ng}(t)$ is the mean and $\boldsymbol{C}_g^x$ variance.

Set $\boldsymbol{\Theta} = (\lambda_1,\lambda_2,\ldots,\lambda_G)$ as unknown parameter vector of the system, $\boldsymbol{A}_n(t) = \left[\mathbb{N}\left(\boldsymbol{X}_n(t)-\hat{\boldsymbol{X}}_{ng}(t),\boldsymbol{C}_g^x\right)\right]_{g\times 1}$ then:

$$\begin{aligned}\boldsymbol{A}_n(t)\cdot\boldsymbol{X}_n(t) &\approx \textstyle\sum_{g=1}^{G}\lambda_g\cdot\mathbb{N}\left(\boldsymbol{X}_n(t)-\hat{\boldsymbol{X}}_{ng}(t),\boldsymbol{C}_g^x\right)\\ &= \boldsymbol{\Theta}\cdot\boldsymbol{A}_n(t)\end{aligned} \tag{11}$$

3.2. Variational Bayesian inference

Variational Bayesian inference is a kind of inference method, which uses simple distribution to approximate the real posterior distribution of unknown variables, and usually assumes that the unknown variables are independent of each other. It is assumed that $\boldsymbol{\Lambda}$ is the hidden variable vector of system state $\boldsymbol{X}$, $\boldsymbol{\Theta}$ is the unknown parameter vector and $\boldsymbol{Y}$ is the observation vector. Variational Bayesian inference has two main purposes: one is to approach the edge likelihood function $p(\boldsymbol{Y}|\boldsymbol{\Lambda},\boldsymbol{\Theta})$ to achieve the model selection, the other is to approach the posterior probability distribution $p(\boldsymbol{\Lambda},\boldsymbol{\Theta}|\boldsymbol{Y})$ including all parameters to achieve the prediction of unknown variables. Assuming that the prior distribution of all parameters in the Bayesian model of unknown variables is known, the purpose of variation is to find a posterior probability distribution to minimize the *KL* divergence (KLD) between it and the real posterior probability distribution.

The principle of variational Bayesian is as follows:

$$\ln p(\boldsymbol{Y}|\boldsymbol{\Theta}) = L(q(\boldsymbol{\Lambda},\boldsymbol{\Theta})) + KL(q(\boldsymbol{\Lambda})\|p(\boldsymbol{\Lambda}|\boldsymbol{Y},\boldsymbol{\Theta})) \tag{12}$$

where, $L(q(\boldsymbol{\Lambda},\boldsymbol{\Theta}))$ represents the lower bound of variation confidence, and $KL(q(\boldsymbol{\Lambda})\|p(\boldsymbol{\Lambda}|\boldsymbol{Y},\boldsymbol{\Theta}))$ represents the differences asymmetric quantity of two probability distributions $q(\boldsymbol{\Lambda})$ and $p(\boldsymbol{\Lambda}|\boldsymbol{Y},\boldsymbol{\Theta})$, namely *KL* divergence, respectively:

$$L(q(\boldsymbol{\Lambda},\boldsymbol{\Theta})) = \ln\int q(\boldsymbol{\Lambda})\frac{p(\boldsymbol{Y},\boldsymbol{\Lambda}|\boldsymbol{\Theta})}{q(\boldsymbol{\Lambda})}d\boldsymbol{\Lambda} \tag{13}$$

$$KL(q\|p) = -\ln\int q(\boldsymbol{\Lambda})\frac{p(\boldsymbol{\Lambda}|\boldsymbol{Y},\boldsymbol{\Theta})}{q(\boldsymbol{\Lambda})}d\boldsymbol{\Lambda} \tag{14}$$

By maximizing the confidence lower bound $L(q(\boldsymbol{\Lambda},\boldsymbol{\Theta}))$ of the variational distribution $q(\boldsymbol{\Lambda},\boldsymbol{\Theta})$, it is equivalent to minimizing the *KL* divergence to obtain the approximation of the posterior probability distribution. That is to say, for any variational distribution $q(\boldsymbol{\Lambda},\boldsymbol{\Theta})$, when the *KL* divergence is 0, the lower bound $L(q(\boldsymbol{\Lambda},\boldsymbol{\Theta}))$ gets the maximum value, and the variational distribution $q(\boldsymbol{\Lambda},\boldsymbol{\Theta})$ is equal to the real posterior distribution $p(\boldsymbol{\Lambda},\boldsymbol{\Theta}|\boldsymbol{Y})$.

3.3. Joint estimation algorithm with sequential variational Bayesian

In this paper, we need to complete the joint estimation of the hidden variables $\boldsymbol{\Lambda}$ of gas distribution state $\boldsymbol{X}$ and sources unknown parameters $\boldsymbol{\Theta}$ and they are assumed independent. According to the system observation model in formula (3), the core idea of the joint optimization method based on the variational Bayesian inference is to use the variational distribution $q(\boldsymbol{\Lambda}_{1:t},\boldsymbol{\Theta}_{1:t})$ to approximate the real posterior distribution of states $p(\boldsymbol{\Lambda}_{1:t},\boldsymbol{\Theta}_{1:t}|\boldsymbol{Y}_{1:t})$, namely:

$$p(\boldsymbol{\Lambda}_{1:t},\boldsymbol{\Theta}_{1:t}|\boldsymbol{Y}_{1:t}) \approx q(\boldsymbol{\Lambda}_{1:t},\boldsymbol{\Theta}_{1:t}) \tag{15}$$

The maximum value of $L(q(\boldsymbol{\Lambda}_{1:t},\boldsymbol{\Theta}_{1:t}))$ can be achieved by the variational Bayesian iterative updating $q(\boldsymbol{\Lambda}_{1:t},\boldsymbol{\Theta}_{1:t})$. By introducing the average field hypothesis, we can decompose it into:

$$q(\boldsymbol{\Lambda}_{1:t},\boldsymbol{\Theta}_{1:t}) = \sum_{i=1}^{t}q(\boldsymbol{\Lambda}_i,\boldsymbol{\Theta}_i) = \sum_{i=1}^{t}q(\boldsymbol{\Lambda}_i)\cdot q(\boldsymbol{\Theta}_i) \tag{16}$$

Optimize each factor $q(\boldsymbol{\Lambda}_i,\boldsymbol{\Theta}_i)$ to achieve the overall optimization. According to the functional theory, the corresponding general solution expression can be obtained by each partial derivation of $q(\boldsymbol{\Lambda}_i,\boldsymbol{\Theta}_i)$.

$$q(\boldsymbol{\Lambda}_i,\boldsymbol{\Theta}_i) = \frac{\exp\left(E_{j\neq i}\left[\ln p(\boldsymbol{Y}_i,\boldsymbol{\Lambda}_i|\boldsymbol{\Theta}_i)\right]\right)}{\int\exp\left(E_{j\neq i}\left[\ln p(\boldsymbol{Y}_i,\boldsymbol{\Lambda}_i|\boldsymbol{\Theta}_i)\right]\right)d\boldsymbol{\Lambda}_i} \tag{17}$$

$E_{j\neq i}[\bullet]$ is the expectation, and it can be obtained with the logarithm of both sides of Eq. (11):

$$\ln q(\boldsymbol{\Lambda}_i,\boldsymbol{\Theta}_i) = E_{j\neq i}\left[\ln p(\boldsymbol{Y}_i,\boldsymbol{\Lambda}_i|\boldsymbol{\Theta}_i)\right] + b \tag{18}$$

where b is defined as an item independent of function arguments,

$$E_{j\neq i}\left[\ln p\left(\boldsymbol{\Lambda}_i\left|\boldsymbol{Y}_i,\boldsymbol{\Theta}_i\right.\right)\right]=\int \ln p\left(\boldsymbol{Y}_i,\boldsymbol{\Lambda}_i\left|\boldsymbol{\Theta}_i\right.\right)\prod_{j\neq i} q\left(\boldsymbol{\Lambda}_i,\boldsymbol{\Theta}_i\right)d\boldsymbol{\Lambda}_i \tag{19}$$

The joint estimation algorithm of Variational Bayesian inference usually involves two stages. In the first VB-E stage, the expectation $\ln q\left(\boldsymbol{\Lambda}_i,\boldsymbol{\Theta}_i\right)$ of each system state $\boldsymbol{\Lambda}_i$ is calculated according to $\boldsymbol{\Theta}_i$ in (18), and the posterior probability distribution is estimated approximately. In the second VB-M stage, the model parameters $\boldsymbol{\Theta}_i$ is updated to maximize the expected value $\ln q\left(\boldsymbol{\Lambda}_i,\boldsymbol{\Theta}_i\right)$ and determine whether the lower bound of the variation reaches the threshold.

3.4. Sequential joint estimation algorithm with Gaussian mixture model

In VB-E stage, the Gaussian mixture probability density function is used to approximate the nonlinear system model to realize the state estimation of gas leakage and diffusion distribution system. Then in the VB-M stage, based on the state estimation results and the environmental observation by the sensor network, the parameters of the Gaussian mixture model are estimated and updated.

VB-E stage: According to formula (3), (15) and (17) update $q\left(\boldsymbol{\Lambda}_{n,i},\boldsymbol{\Theta}_{n,i}\right)$ firstly, it can be described as follows:

$$\begin{aligned} L\left(q\left(\boldsymbol{\Lambda}_{n,i},\boldsymbol{\Theta}_{n,i}\right)\right) &= E_{j\neq i}\left[\ln p\left(\boldsymbol{\Lambda}_{n,i},\boldsymbol{\xi}_{n,i}\left|\boldsymbol{Y}_{n,i},\boldsymbol{\Theta}_{n,i}\right.\right)\right]_{q\left(\boldsymbol{\Theta}_{n,i}\right)} \\ &\propto -\frac{1}{2}\left(\boldsymbol{\Lambda}_{n,i}-\boldsymbol{u}_{n,i}\right)^T \boldsymbol{C}_{n,i}{}^{-1}\left(\boldsymbol{\Lambda}_{n,i}-\boldsymbol{u}_{n,i}\right) \end{aligned} \tag{20}$$

So, $q\left(\boldsymbol{\Lambda}_{n,i},\boldsymbol{\Theta}_{n,i}\right)$ is a Gaussian distribution with mean $\boldsymbol{u}_{n,i}$ and variance $\boldsymbol{C}_{n,i}$, where mean $\boldsymbol{u}_{n,i}$ and variance $\boldsymbol{C}_{n,i}$ are respectively expressed as:

$$\boldsymbol{C}_{n,i}=\left(\frac{1}{b}\boldsymbol{A}_{n,i}^T\boldsymbol{A}_{n,i}+\boldsymbol{C}_{n,0}^{-1}\right)^{-1} \tag{21}$$

$$\boldsymbol{u}_{n,i}=\frac{1}{b}\boldsymbol{C}_{n,i}^{-1}\boldsymbol{A}_{n,i}^T\left(\boldsymbol{Y}_{n,i}-\boldsymbol{\Theta}_{n,i}E\left(\xi_{n,i}\right)_{q\left(\boldsymbol{\Theta}_{n,i}\right)}\right) \tag{22}$$

Secondly, $q\left(\boldsymbol{\xi}_{n,i}\right)$ is updated and estimated approximately as follows:

$$\begin{aligned} L\left(q\left(\boldsymbol{\xi}_{n,i}\right)\right) &= E_{j\neq i}\left[\ln p\left(\boldsymbol{\Lambda}_{n,i},\boldsymbol{\xi}_{n,i}\left|\boldsymbol{Y}_{n,i},\boldsymbol{\Theta}_{n,i}\right.\right)\right]_{q\left(\boldsymbol{\Lambda}_{n,i}\right)} \\ &\propto \left\{-\frac{1}{2b}\left\|\boldsymbol{Y}_{n,i}-\boldsymbol{\Theta}_{n,i}\boldsymbol{\Lambda}_{n,i}\right\|_2^2\right\} \end{aligned} \tag{23}$$

So, $q\left(\boldsymbol{\xi}_{n,i}\right)$ is a truncated Gaussian distribution, and the $E\left(\boldsymbol{\xi}_{n,i}\right)_{q\left(\boldsymbol{\Theta}_{n,i}\right)}$ in Eq. (22) can be calculated as follows.

At first, the posterior probability density function of $\boldsymbol{\xi}_{n,i}$ can be obtained from the conditional probability formula:

$$p\left(\boldsymbol{\xi}_{n,i}\left|\boldsymbol{Y}_{n,i},\boldsymbol{\Theta}_{n,i}\right.\right)=\frac{p\left(\boldsymbol{Y}_{n,i},\boldsymbol{\Theta}_{n,i}\left|\boldsymbol{\xi}_{n,i}\right.\right)p\left(\boldsymbol{\xi}_{n,i}\right)}{\int p\left(\boldsymbol{Y}_{n,i},\boldsymbol{\Theta}_{n,i}\left|\boldsymbol{\xi}_{n,i}\right.\right)d\boldsymbol{\xi}_{n,i}} \tag{24}$$

$$\int p\left(\boldsymbol{Y}_{n,i},\boldsymbol{\Theta}_{n,i}\left|\boldsymbol{\xi}_{n,i}\right.\right)d\boldsymbol{\xi}_{n,i}=\int\frac{1}{\sqrt{2\pi b}}\exp\left(-\frac{\left(\boldsymbol{Y}_{n,i}-\boldsymbol{\Theta}_{n,i}\boldsymbol{\Lambda}_{n,i}-\boldsymbol{\xi}_{n,i}\right)^2}{2b}\right)d\boldsymbol{\xi}_{n,i} \tag{25}$$

Assumed $\boldsymbol{u}_\xi=\boldsymbol{Y}_{n,i}-\boldsymbol{\Theta}_{n,i}\boldsymbol{\Lambda}_{n,i}$, we can get:

$$\int p\left(\boldsymbol{Y}_{n,i},\boldsymbol{\Theta}_{n,i}\left|\boldsymbol{\xi}_{n,i}\right.\right)d\boldsymbol{\xi}_{n,i}=cdf\left(\boldsymbol{u}_\xi\right)-cdf\left(\boldsymbol{I}_\xi\right) \tag{26}$$

where $cdf(\bullet)$ represents the cumulative distribution function of standard normal distribution.

$$cdf\left(\boldsymbol{u}_\xi\right)=\int\frac{1}{\sqrt{2\pi}}\exp\left(-u_\xi^2/2\right)d\boldsymbol{u}_\xi \tag{27}$$

Then

$$p\left(\boldsymbol{\xi}_{n,i}\left|\boldsymbol{Y}_{n,i},\boldsymbol{\Theta}_{n,i}\right.\right)=\frac{1}{cdf\left(\boldsymbol{u}_\xi\right)-cdf\left(\boldsymbol{I}_\xi\right)}p\left(\boldsymbol{Y}_{n,i},\boldsymbol{\Theta}_{n,i}\left|\boldsymbol{\xi}_{n,i}\right.\right) \tag{28}$$

then

$$\begin{aligned} E\left(\boldsymbol{\xi}_{n,i}\right)_{q\left(\boldsymbol{\Theta}_{n,i}\right)} &= \frac{1}{cdf\left(\boldsymbol{u}_\xi\right)-cdf\left(\boldsymbol{I}_\xi\right)} \\ &\cdot\int\boldsymbol{\xi}_{n,i}\cdot p\left(\boldsymbol{Y}_{n,i},\boldsymbol{\Theta}_{n,i}\left|\boldsymbol{\xi}_{n,i}\right.\right)p\left(\boldsymbol{\xi}_{n,i}\right)d\boldsymbol{\xi}_{n,i} \end{aligned} \tag{29}$$

VB-M stage: Substituting $q\left(\boldsymbol{\Lambda}_{n,i},\boldsymbol{\Theta}_{n,i}^{old}\right)$ in $L\left(q\left(\boldsymbol{\Lambda}_{n,i},\boldsymbol{\Theta}_{n,i}\right)\right)$, the parameters vector $\boldsymbol{\Theta}_{n,i}$ could be estimated by the following formula:

$$\begin{aligned} \hat{\Theta}_{n,i} &= \arg\max E_{j\neq i}\left[\ln p\left(\boldsymbol{\Theta}_{n,i}\left|\boldsymbol{Y}_{n,i},\boldsymbol{\xi}_{n,i}\right.\right)\right]_{q\left(\boldsymbol{\Lambda}_{n,i},\boldsymbol{\Theta}_{n,i}^{old}\right)} \\ &\propto -\frac{1}{2}\ln\left(\left|\boldsymbol{C}_{n,0}\right|\right)-\frac{1}{2}\boldsymbol{Y}_{n,i}^T\boldsymbol{C}_{n,0}^{-1}\boldsymbol{Y}_{n,i} \\ &\propto -\ln\left(\prod_{g=1}^{g}\lambda_g\right)-\frac{1}{2}tr\left(\boldsymbol{C}_0^{-1}\left(\boldsymbol{C}_{n,i}+\boldsymbol{u}_{n,i}\boldsymbol{u}_{n,i}^T\right)\right) \end{aligned} \tag{30}$$

Derivative formula (30) and set the derivation to zero:

$$\frac{\partial\hat{\boldsymbol{\Theta}}_{n,i}}{\partial\lambda_g}=\frac{M}{2\lambda_g}+\frac{M}{2\lambda_g}tr\left(\boldsymbol{C}_0^{-1}\left(\boldsymbol{C}_{n,i}+\boldsymbol{u}_{n,i}\boldsymbol{u}_{n,i}^T\right)\right)=0 \tag{31}$$

It could be get:

$$\hat{\boldsymbol{\Theta}}_{n,i}=\left(\hat{\lambda}_g=\frac{tr\left(\boldsymbol{C}_0^{-1}\left(\boldsymbol{C}_{n,i}+\boldsymbol{u}_{n,i}\boldsymbol{u}_{n,i}^T\right)\right)}{M}\right)_{1\times M} \tag{32}$$

4. Adaptive grid division method for gas source localization

It can be seen from the above discussion that the grid division method of the monitoring area affects the performance of the target source localization. If the grid division size is too small, the under determined linear equations in formula (7) will increase sharply, which will greatly increase the calculation of reconstruction algorithm; if the size of grid division is too large, there may be multiple targets in one grid, and the target is easy to be lost during the localization process. Therefore, an adaptive grid division size can improve the positioning accuracy and reduce the computational complexity of the system. In this paper, two grid division methods are used to locate the gas release sources in different stages, it was named as the global large-scale grid division and local small-scale grid division method, and the specific process is as follows:

(1) Global large scale grid division and localization

In the initial stage of the gas release source localization in sensor networks, the monitoring area is usually divided into large scales globally in order to reduce the signal sensing overhead, and the candidate grids within the target source existing is initially predicted. Assuming that the monitoring area is divided into N_0 initial grids, the initial grid state vector $\boldsymbol{\Lambda}$ was given to describe the gas source distribution state $\boldsymbol{X}$ and it is defined as $\boldsymbol{\Lambda}_0=\left(\boldsymbol{r}_0,\boldsymbol{\gamma}_0\right)$, where $\boldsymbol{r}_0$ is the center position vector of the initial grid, $\boldsymbol{\gamma}_0$ is the vector to describe whether there is a target in the grid, $\gamma_{n,0}=1$ or $\gamma_{n,0}=0$ respectively represents whether there is or not exist a target source in the initial grid. According to the principle of compressed sensing, M nodes are randomly selected in the region for target compressed sensing, and the observations are transmitted to cluster head nodes for the state vector $\boldsymbol{\Lambda}$ reconstruction. The estimated values of reconstruction errors are calculated based on VB-E step with the formula (29), the reconstruction values of sparse vectors $\boldsymbol{\Lambda}_0=\left(\boldsymbol{r}_0,\boldsymbol{\gamma}_0\right)$ would be get.

(2) Local scale adaptive grid division and localization

Assuming that in the process of global large-scale localization, a total number L of target grids are selected, and local scale adaptive grid division and localization is aimed at the $l\in L$ grid, in which, the initial $l\in L$ grid would be re-divided to further determine the exact location of the target with adjusting the size adaptively, and the number of grid division depends on the number value $\gamma_{l,i}$ in $\boldsymbol{\Lambda}_i=\left(\boldsymbol{r}_i,\boldsymbol{\gamma}_i\right)$. The cluster head node would calculate the estimation value of reconstruction error

and gives the estimation value $\boldsymbol{\Lambda}_i = (\boldsymbol{r}_i, \boldsymbol{\gamma}_i)$. If $\gamma_{l,i} > 1$, there are still multiple target sources in one grid, the grid will be re-divided and the new observation value will be received for updating; Otherwise, it will be stopped. According to formula (32), the optimal sparse vector $\hat{\boldsymbol{\theta}}_{n,i}$ reconstruction value is given by cluster head based on VB-M step.

At the initial stage of the proposed adaptive grid division and source localization method, the global large-scale grid division and the centralized information processing method is used to complete the screening of the existing and non-existent grids in the monitoring area. In the subsequent stage, the local small-scale adaptive grid division method is used to further divide the different candidate grids, and the distributed parallel information processing method is used to accurately locate the target source in the candidate grid regions. This method is more easily applied to the current hierarchical structure of sensor network information processing applications and extended to multi-target positioning problems in different scenarios, which provides a new idea for high-precision and high-efficiency multi-target positioning with the combination of the global and local division, centralized and distributed processing, coarse and fine positioning method.

According to the above discussion, the proposed gas release source location algorithm based on hierarchical adaptive grid division method in this paper is shown in Fig. 1.

It is mainly based on the Bayesian expectation maximization joint estimation algorithm and the adaptive grid division strategy be described as follows: (1) firstly, the monitoring area is divided by the global large-scale grid division method and the condition of the system variables was initialized, then the compressed sensing matrix is constructed and the environment information was compressed sensed, and the hidden variable $\boldsymbol{\Lambda}_i = (\boldsymbol{r}_i, \boldsymbol{\gamma}_i)$ estimation of the candidate grid is given based on the Bayesian joint estimation method (2) According to the candidate grid and the variance of the estimator, the candidate grid is further divided by local small scale, and with the local environment information sampling by the cluster nodes, the sparse reconstruction of the hidden variables and unknown parameters of the gas release source distribution is completed based on the joint estimation algorithm of the variational Bayesian; (3) At each stage, the variance of the system estimator is determined whether to meet the threshold to control the execution and end of the algorithm.

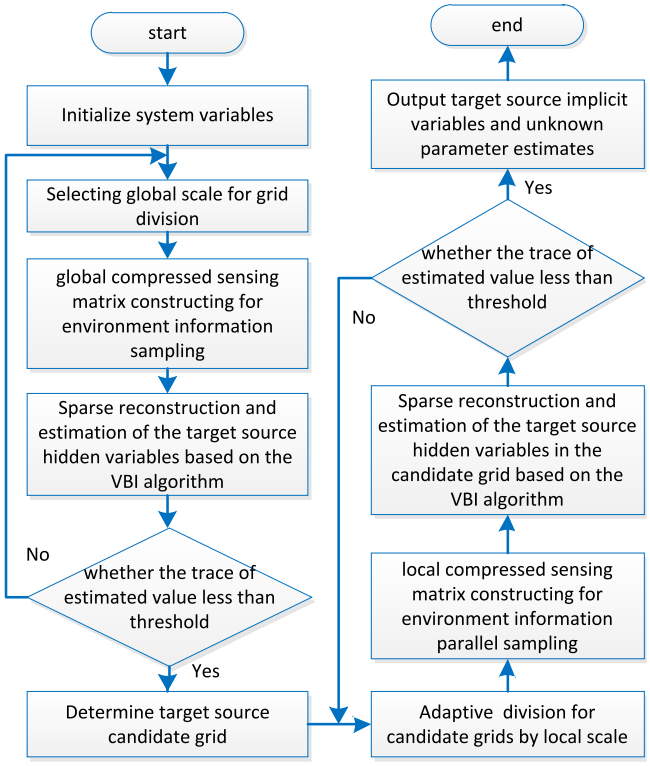


Fig. 1. The joint gas source state and parameters estimation algorithm based on VBI with adaptive grid division method.

5. Simulations and analysis

5.1. Simulation environment parameters

In this paper, the simulation of proposed algorithm is fulfilled on the MATLAB platform, assuming that the coordinate position information of each node in the sensor network can be obtained by other methods, and each node can complete the real-time information collection in the whole simulation process. The collected information has the characteristics of space–time consistency. Each measurement value adds a Gaussian white noise with the mean value of 0 and the variance of $\lambda = \sigma^2$, and defines the calculation method of signal-to-noise ratio is $SNR = 10\lg\left(\|\boldsymbol{AX}\|_2^2 / M \cdot |C|\right)$.

In this paper, the performance evaluation indexes of gas source parameter estimation algorithm in sensor network mainly include:

(1) Mean localization accuracy rate (*MLAR*).

If the location algorithm finds the reconstruction point $\hat{\boldsymbol{r}}_s$ within the range of less than 1 grid from the real target $\boldsymbol{r}_s$, it is considered that the target is successfully localized, and the ratio of the number of successfully localized targets (or grid number) to the total number of targets is recorded as the average location accuracy.

$$MLRA = \frac{\hat{K}}{K}, \quad \hat{K} = \frac{1}{N}\sum_{i=1}^{N} K_i$$

(2) The estimation accuracy is mainly described by mean estimation accuracy (*MEA*), which is defined as the average Euclidean distance between the real position of the target and the position of the nearest recovery point. The formula is as follows:

$$MEA = \frac{1}{\hat{K}}\sum_{i=1}^{\hat{K}} \|\hat{\boldsymbol{r}}_s - \boldsymbol{r}_s\|$$

where the coordinates $\boldsymbol{r}_s$ representing the real position of the target and the coordinates $\hat{\boldsymbol{r}}_s$ representing the estimated position of the target.

(3) The Mean run time in order to evaluate the location performance, the multi-target location time is defined as

$$MRT = T_{globle} + T_{local}$$

where T_{globle} is the localization time of global large-scale grid division and T_{local} is the localization time of local small-scale grid division.

5.2. The algorithm performance analysis

The simulation of the proposed multi-target source localization algorithm based on joint sparse estimation method with hierarchical adaptive grid method was given in this paper. The performance of different global grid scale division and localization and the performance of source localization with different compressed sensing algorithms were analyzed, the results showed that the proposed algorithm has better localization performance.

5.2.1. The performance of different global scale grid division and localization

The global large-scale grid division sizes affect the setting of the adaptive grid and the performance of source localization. The following simulations analyze the performance of different large-scale division sizes. The targets and nodes are randomly deployed in the 200 m*200 m monitoring area, and the average localization time is defined as the

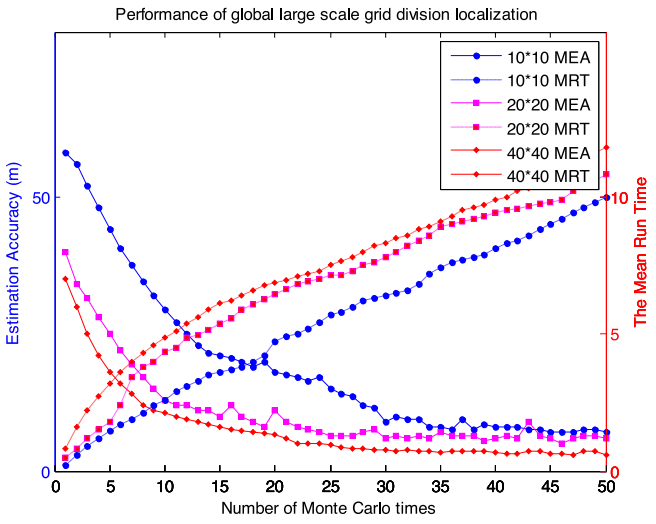


Fig. 2. The performance of different global scale grid division and localization.

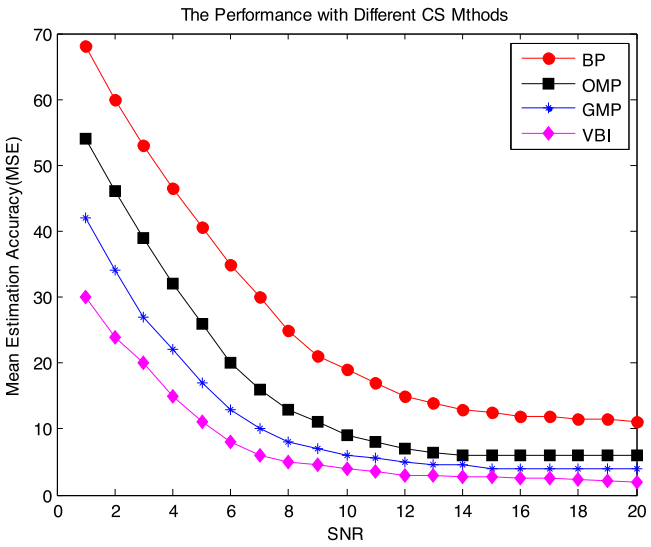


Fig. 3. The mean estimation accuracy of different compressed sensing algorithms with different SNRs.

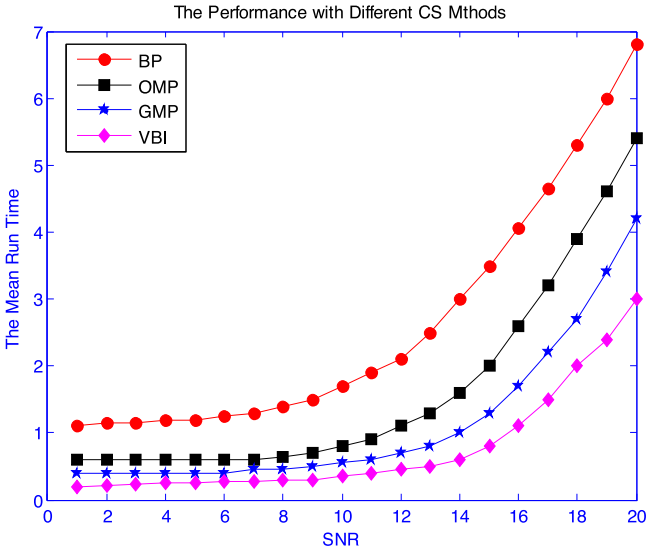


Fig. 4. The mean run time performance of different compressed sensing algorithms with different SNRs.

evaluation index:

$$MTR_{globle} = \frac{1}{N_{globle}} \sum_{i=1}^{N_{globle}} t_i$$

where $N_{globle} = 50$ represents the number of Monte Carlo experiments, t_i is the multi-target positioning time of the ith experiment.

For different global large-scale grids division sizes: 10*10, 20*20, 40*40, the results was shown in Fig. 2, it can be seen that the average accuracy of target source location will continue to improve with the increase of grid number, and the estimation error will continue to decline, but when the initial grid number reaches a certain scale, both of them will tend to be stable. It is also shown that the global grid scale setting is too large to get accurate system positioning in a short time when the monitoring area is divided with fixed grid sizes.

5.2.2. The performance of different compressed sensing reconstruction algorithms

The performance of different compressed sensing reconstruction algorithms in 200 m*200 m monitoring area was analyzed. In which, the traditional compressed sensing positioning algorithm directly divides the monitoring area into 20*20 grid size and reconstructed the K targets vector with the BP algorithm, OMP algorithm or GMP algorithm. As for the proposed method in this paper, the algorithm first divided the monitoring area into 10*10 grids in the initial global grid division stage, and then uses adaptive grid division method for the candidate grid to locate the target source and estimate the exact location of K targets.

In order to analyze the performance of the different CS algorithms with different SNRs, there are $K = 7$ targets and m = 50 sensor nodes randomly deployed in the monitoring area. Fig. 3 shows that with the increase of signal-to-noise ratio, the target signal strength received by the node is significantly improved, so the average localization error of the four CS algorithms shows a downward trend, and the positioning accuracy of the VBI algorithm in this paper is significantly better than the other three algorithms. The average localization error of different four compressed sensing reconstruction algorithms with the SNR changes from 1 dB to 20 dB was shown in Fig. 3, it could be seen that the performance of the VBI algorithm in this paper is better than the BP, OMP and GMP algorithms. When the SNR is greater than 10 dB, the performance of the four algorithms tends to be stable. The accuracy of the VBI algorithm in this paper can reach within one grid, while the other three kinds of BP, OMP and GMP algorithms are within 10, 6 and 2 grids respectively. In Fig. 4, it could be seen that the average positioning time of the four CS algorithms increases gradually with the increase of signal-to-noise ratio, but the average localization time of the VBI algorithm in this paper is significantly less than that of BP, OMP and GMP algorithm, so the performance of the VBI algorithm in this paper is better than the other traditional compression sensing positioning algorithm.

6. Conclusions

In this paper, a joint estimation method based on variational Bayesian inference algorithm for gas source location is proposed with a hierarchical adaptive grid division strategy. In the initial global large-scale grid division and localization stage, the VBI algorithm was mainly used for the joint sparse estimation of the target source distribution and the initial candidate grid reconstruction. In the local adaptive scale grid division and localization stage, the sparse reconstruction of target source unknown parameters in candidate grid is realized by the distributed variational Bayesian inference method with the adaptive

grid re-divided method, and Simulation results show that the proposed method could achieve higher performance than traditional compression sensing methods, and balance the contradiction between algorithm performance and network energy consumption.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

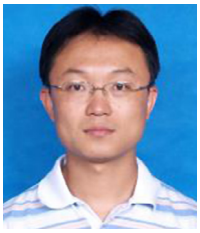
Zhang Yong: Methodology. **Wang Tong:** Formal analysis. **Shi Yu:** Conceptualization. **Zhang Liyi:** Data curation.

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